6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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1 Introduction

Clouds are a major component in regulating the Earth's radiative budget and the hydrological cycle. They interact with the solar and thermal radiation affecting the energy balance and the extent of this impact depends mainly on the cloud's physical properties Twomey; Brenguier et al.; Zeng et al.. However, these properties vary significantly, making the study of clouds quite complex. Although gases and aerosols show considerable spatial and temporal variability, clouds have even greater variability. (int). Additionally, the phase of hydrometeors within the clouds can vary significantly, affecting the cloud radiative properties (Yoshida and Asano).

Partly because of this, clouds remain poorly understood. The lack of data in climatically sensitive regions, such as the Mediterranean basin (Marinou et al.) also contributes to this gap of knowledge about clouds. Satellite studies using active remote sensors provide a general picture of cloud properties, but they present low temporal resolution and information on low level clouds is affected by large uncertainties. Additionally, this data should be validated with the use of ground-based remote sensors such as lidars and cloud radars (Protat et al., a, c). To overcome these difficulties obtaining more detailed information, ground-based active remote sensors observations are required. There are studies on clouds using ground-based remote sensing measurements in Europe, but they are mostly located in Northern Europe (Stefan Kneifel et al.; Tatiana Nomokonova et al.; Bühl et al.; Peggy Achtert et al.).

However, the information on the Mediterranean region is scarce. Clouds in this region form through various mechanisms, including thermodynamic conditions, Sahara dust intrusions (Casquero-Vera et al.), and cut-off lows (COLs) systems . These events are related to convection, increasing in CCN concentration (dust) and cold air masses at high altitudes that were separated from the jet stream, respectively. These complex formation mechanisms pose significant challenges for accurately representing these clouds in atmospheric models. Consequently, there is a critical need for enhanced cloud analysis and data collection in the Mediterranean region. Since 2018, the AGORA (Andalusian Global ObseRvatory of the Atmosphere) ACTRIS CCRES station, in the southeast of the Iberian Peninsula, located in the western Mediterranean basin, has been providing cloud remote sensing measurements. This dataset represents a database for cloud statistics, providing comprehensive data for cloud analysis over an extended period. Notably, it stands as the only cloud remote sensing dataset available in Spain, making it an invaluable resource for regional climate studies. It contributes significantly to filling the gap in cloud data for this climate variability hotspot, enhancing our understanding of cloud dynamics and aiding in the improvement of predictive models and climate studies in these critical areas

This study aims to classify different cloud categories observed over the AGORA station and characterize their physical properties (microphysics and macrophysics). Cloudnet data is used to classify clouds into distinct types based on the percentage of different hydrometeors they contain, allowing us to analyze their physical properties separately. Particularly, we focus on assessing a new approach for cloud classification and evaluate its impact on the derived cloud properties. Several studies have explored cloudnet data to perform cloud statistics through profile-based cloud classification. At the Ny-Ålesund station in the Arctic, this method was employed to characterize single layer clouds and analyze their monthly variability during two years (Tatiana Nomokonova et al.). Similarly, in the German alps, Stefan Kneifel et al. used the same method to classify ice clouds in order to evaluate cloud variability at high altitude locations. Lately, Răzvan Pîrloagă et al. applied a methodology quite similar to that of Tatiana Nomokonova et al. for cloud classification. Then, seasonal cloud variability was achieved using multi-year Cloudnet data in Bucharest, Romania. Conversely, this study proposes a novel cloud characterization approach, based on hydrometeor cluster classification. This new method marks an advancement over previous studies, avoiding high correlation between cloud type statistics. It was applied in one of the Southernmost Cloudnet stations inEurope, specifically in the western mediterranean.

The experimental site, datasets, instruments and products are described in Section 2. The proposed new approach on cloud classification algorithm and the cloud assessment are presented in Section 4. Section 5 analyzes the annual, seasonal, and daily cycle of cloud types, as well as the base and top height, and thickness. In addition, this section discusses the liquid and ice properties variability for different clouds. Finally, Section 6 summarizes the main findings and discusses their importance to improve further cloud statistics analysis.

2 Experimental site and Instrumentation

The AGORA ACTRIS-CCRES station (37°2’N, 3°6’E, 680 masl) is located in the South of the Iberian Peninsula being one the most meridional CLOUDNET stations. This station is characterized by a continental climate, with colder months during winter and spring and warm months during summer and fall. Especially between summer and winter,temperature and humidity present the largest contrast, as revealed by Andrés Esteban Bedoya-Velásquez et al.. The authors show that these changes are mainly due to solar incidence, being the diurnal cycle a strong component in seasonal differences. For non cloudy data, typically surface temperature in summer can vary around 30 – 40 °C, meanwhile they vary around 15 °C in winter. The relative humidity below 5 km a.g.l presents the opposite behavior, ranging between 60-75% during winter, and 40-50% during summer. These thermodynamic conditions will be particularly important to understand diurnal cycle of cloud types and inter-annual differences in cloud properties analyzed in further sections.

2.0.1 Instrumentation

In this study we use a combination of active and passive remote sensors that are part of ACTRIS CCRES and exploited in CLOUDNET. These instruments include a microwave radiometer, a ceilometer and a cloud radar. A synergistic combination of these instruments have been widely used for many Cloudnet stations. This combination shows a promising configuration for long-term cloud observations (J. A. et al.), especially to calculate cloud microphysics, providing better accuracy than satellites (Protat et al., b).

The RPG-HATPRO G2 (Humidity and Temperature PROfiler) MWR is a passive remote sensing instrument that measures the brightness temperature around the water vapor (22-31 GHz) and oxygen (51-58 GHz) absorption bands. This instrument has radiometric accuracy of 0.3-0.4 K and provides accurate values of retrieved liquid water path (LWP) and integrated water vapor (IWV) with 1 s of integration time (Löhnert and Crewell). In addition, it provides temperature and humidity profiles with the same time resolution. More details can be found in Francisco Navas-Guzmán et al.; Rose et al.

The CHM15k Nimbus ceilometer is an active remote sensing instrument that works with a pulsed laser of Nd:YAG. It measures in the photon-counting mode and provides the attenuated backscattering coefficient (β) of aerosols and small cloud droplets. This instrument has repetition frequency range about 5-7 kHz, where each pulse is emitted at 1064 nm with 8.4 μJ of energy. The laser beam has a range resolution of 15 m and can diverge 0.3 mrad, reaching a complete overlap of about 1500m a.g.l (Heese et al.). Especifically, this ceilometer is operated with a time resolution of 15 s (A. Cazorla et al.). The dual polarimetric RPG-FMCW 94 GHz Doppler cloud radar is also an active remote sensing instrument that operates at 94 GHz. This instrument uses the frequency-modulated continuous wave (FMCW) signal which makes it possible to vary the vertical and temporal resolution. Their frequency range is from 300-3700 kHz and the antenna beam width is 0.56 °. It measures the Doppler velocity spectrum for horizontal and vertical linear polarizations at 94 GHz. Unlike traditional single polarization measurements that retrieve only reflectivity (Z), mean Doppler velocity (ν_D), and spectral width (S_ω), this advanced radar also derives the linear depolarization ratio (LDR), enabling the detection of the melting layer, aerosols, and insects as described by Hogan and O'Connor. Additionally, the instrument was mounted on a platform that enabled it to perform complex scans at different geometries. Table 1 provides an overview of the AGORA instrumentation, detailing operational frequencies or wavelengths, type of measurements, products, measurement ranges, data collection intervals, and references with detailed descriptions.

Table 1. Summary of instruments and their data products.

Instrument	Bands / wavelengths	Measured	Products	Range	Time	References
94 GHz Cloud Doppler Radar	94 GHz (W-band)	Doppler velocity spectrum (DVS)	ν_D , Z, S_w	[13,51] m	[1.1, 3.6] s	Nils Küchler et al.
MWR RPG HATPRO	22-31 GHz (K-band) 51-58 GHz (V-band)	Brightness Temperature	q and T LWP	[10, 300] m -	[2, 2.5] min 2 s	Francisco Navas-Guzmán et al.
CHM15k Nimbus Ceilometer	1064 nm		β	15 m	15 s	A. Cazorla et al.

This range varies with the height and depends on the chirp table configuration.

It also depends on the chirp type.

Humidity and temperature profiles have a range varying from 10 to 150 m in the first 4.4 km and from 50 to 300 m between 4.4 and 10 km.

3 Cloudnet database

The AGORA station has been part of Cloudnet since 2018 and was integrated into ACTRIS CCRES (where ACTRIS stands for Aerosols, Clouds, and Trace gasses Research InfraStructure, and CCRES for Centre for Cloud Remote Sensing) in 2023 as a new national facility. The Cloudnet processing chain (J. A. et al.) performs extensive quality assurance on raw instruments for providing high-level products, such as target classification, liquid water path, and droplet effective radius. This chain standardizes data processing within the cloud remote sensing community. Their products are derived using a synergistic approach, integrating vertically pointing measurements from ground-based instruments like Doppler cloud radars, microwave radiometers, and ceilometers. Specifically, the target classification product, generated through these three instruments, has been utilized in numerous studies for cloud typing and multi-year analysis (Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.). Therefore, an extensive dataset spanning the years 2018 to 2023 was utilized in this study. Figure 1 illustrates the monthly Cloudnet high level products availability for the AGORA ACTRIS-CCRES station. The availability of this data depends on simultaneous vertically pointing measurements of the microwave radiometer, ceilometer, Doppler cloud radar, as well as the meteorological model products - the ECMWF model (European Centre for Medium-Range Weather Forecasts). As observed, the months with lower data availability are Jan-Mar, with 40-50% of the data available, whereas from Apr to Oct the data availability is over 60% .The years 2018 (270,000 profiles) and 2020 (968,769 profiles) are the years with the least and greatest amount of data. The white bars denote periods with missing data due to instrument maintenance, technical issues, and scanning measurements (which are not processed by Cloudnet). Despite the missing periods, the recorded dataset establishes a solid database, with a large enough sample to perform the analysis of cloud cover above AGORA.

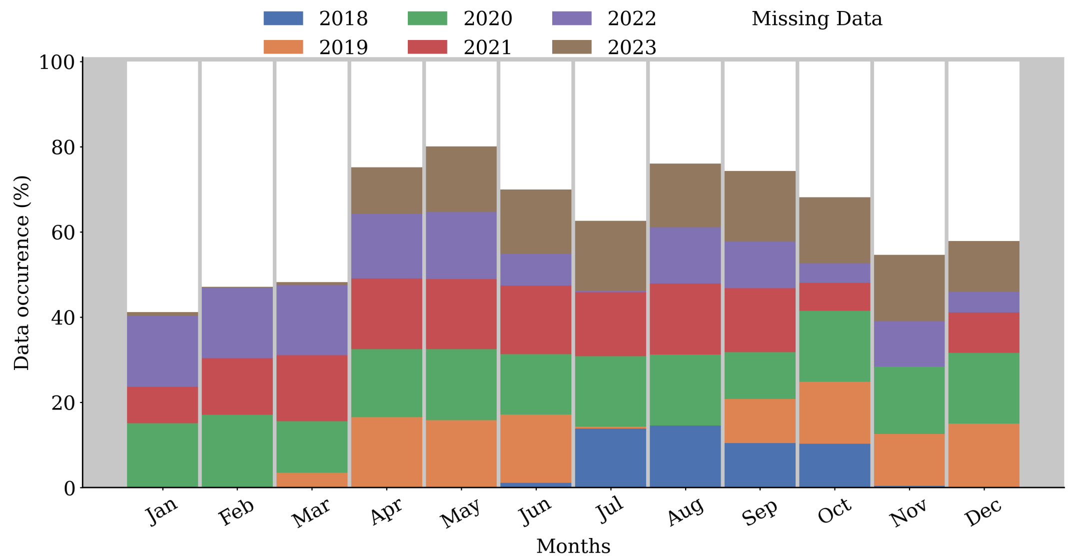

Figure 1. High level post-processed data availability per month at the AGORA ACTRIS CCRES station. This dataset represents more than half a decade of synergic ground-based measurements at this station. Each color represents the data availability for different years.

3.0.1 Cloudnet products

The Cloudnet products used in this study include target classification and microphysical retrievals, such as effective radius, and liquid and ice water content. These were re-gridded to a temporal resolution of 30 seconds and spatial resolution determined by the radar chirp table (Table 1). The cloud radar reflectivity was corrected for two-way gas and liquid attenuation, crucial for high-frequency radars, which are notably attenuated by gases and liquid droplets (Hogan et al.). Data from cloud radar, ceilometer, microwave radiometer, and the ECMWF model were used to derive the target classification (Hogan and O'Connor). This classification assigns integer values (0-10) to atmospheric particles by height and time, with categories ranging from Clear sky (0), Droplets (1), Drizzle or rain (2), Drizzle & droplets (3), Ice (4), Ice & droplets (5), Melting ice (6), Melting & droplets (7), Aerosol (8), Insects (9), Aerosol & insect (10). Liquid and supercooled droplets were identified using lidar attenuated backscatter and radar Doppler velocity spectra (Schimmel et al.), applied only at temperatures above -40°C. Rain was identified as falling liquid particles below the cloud base, while ice was attributed to particles with wet-bulb temperatures below freezing. The melting layer was detected using wet-bulb temperature divergence and linear depolarization ratio from cloud radar. Microphysical properties were retrieved for hydrometeor pixels classified by the target classification. Liquid water content (LWC) and droplet effective radius (r_{liq}) were retrieved in liquid water pixels, including those containing ice and droplets. LWC retrieval, scaled to match liquid water path (LWP) from the microwave radiometer, has a 15% relative uncertainty (A. S. Frisch et al.). Droplet effective radius retrievals, based on a size distribution method, are also subject to 15% uncertainty due to reflectivity errors (S. Frisch et al.). Input data and uncertainties are detailed in Table 2 For ice pixels, ice water content (IWC) was retrieved using the Z-IWC-T relation based on radar reflectivity and temperature (Robin J. Hogan et al.). The W-band retrievals have a relative error ranging from +55% to -35% for temperatures between -20°C and -10°C, and +90% to -47% for temperatures below -40°C. The main source of error is reflectivity, where a 3% reflectivity error can lead to a 12% error in ice water path (IWP) (Pablo Saavedra Garfias et al.). The ice effective radius (r_{ice}) was retrieved using the Z-T relation, with a 50% relative uncertainty (Hannes Griesche et al.), highlighting the challenges of accurately retrieving ice properties (Table 2).

Table 2. Cloudnet microphysics products used in this study, their inputs and assumptions

Cloudnet products	**Input role**	σ_{re}/r_e	**References**
Liquid water content (LWC)	Z and β: cloud base and top identification T, P, assuming an adiabatic lifting MWR or CDR LWP to scale LWC	15%	Frisch et al. (1998)
Ice water content (IWC)	Z and T profiles	-35% to 90%	Robin J. Hogan et al. Julien Delanoë et al.
Droplet effective radius (r_{liq})	Z, and N and σ of lognormal PSD	15%	S. Frisch et al.
Ice effective radius (r_{ice})	Z and T	50%	Hannes Griesche et al.

4 Methodology

4.1 Cluster classification method

The cluster-based algorithm classifies groups of hydrometeors into cloud types by using a cloud mask from the target classification product to identify connected hydrometeors in height and time. These clusters are then categorized into cloud types based on the percentage of different hydrometeors they contain. Unlike profile-based methods, which classify clouds column-by-column and often result in abrupt changes in cloud classification over short time intervals, the cluster-based approach aims to provide a more continuous, time-homogenized classification. This method has proven effective in reducing high correlations between ice and mixed-phase cloud properties, offering a more consistent cloud classification across time.

In both cases, the Cloudnet target classification product is used as a basis for cloud typing. This product provides the hydrometeor type information for each profile. Thus, the pixels classified as hydrometeors, such as Droplets, Ice and Drizzle & Droplets are used to classify clouds. In the case of the profile-based method, Bühl et al. already adapted it for only mixed phase clouds, selecting only cases with more than 15 min of coherent cloud profiles (with similar cloud profiling within 300 s). Nevertheless, by adding new criterias to identify liquid and ice clouds, this method can classify segments of one cloud in different categories, depending on the time threshold. Hence, we introduce a new approach that does not classify cloud profiles neither segments, but cloud clusters.

Prior to the use of the target classification product, a preprocessing of the data was required. It has been observed that for some specific cases, misclassification of some pixels may occur. Figure 2.b illustrates a thinand lower precipitable ice cloud present between 08:00 and 20:00 at 2 km (turquoise layer in Fig 2.a). However, there is no evidence of clouds in the attenuated backscatter (Fig 2.d) and the radar reflectivity (Fig 2.c). These misclassifications stem from an inaccurate representation of the temperature profile by the models in the region of Granada, affecting the target classification algorithm, and subsequently, the cloud classification performed in this study. To mitigate these issues and prevent misinterpretation of future cloud statistics, we established a filter to detect and remove these cases. Basically, all these precipitating and non precipitating ice clouds which were identified with mean cloud base below 4 km (above mean sea level) and average cloud thickness below 700 meters were reclassified as noise. These values were empirically determined by analyzing many different days of target classification products at the Granada station

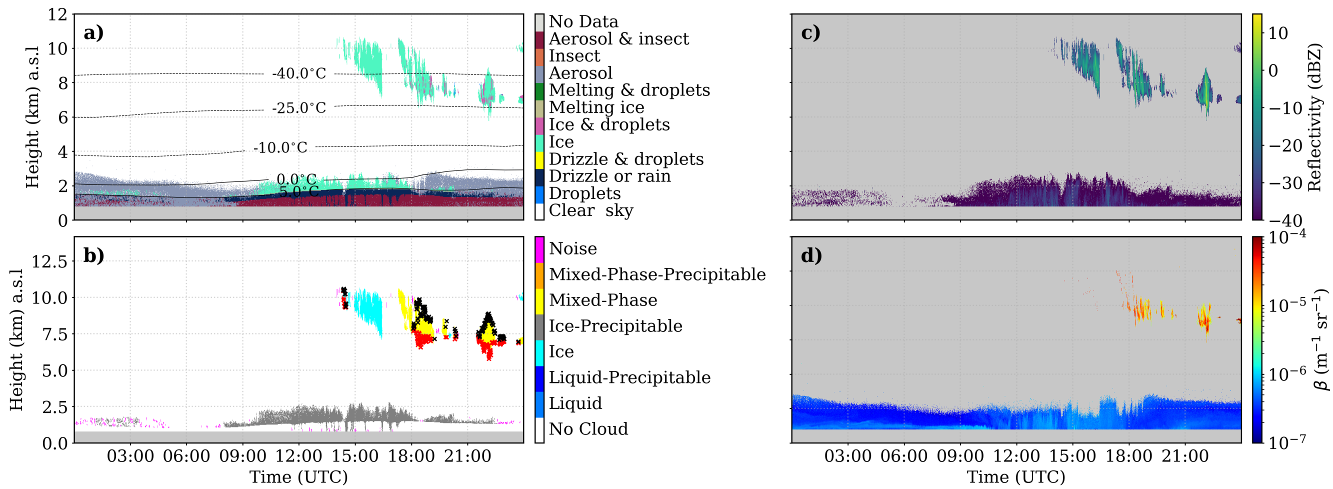

Figure 2. A case study of Cloudnet hydrometeor misclassification. (a) Target classification product; (b) Radar reflectivity factor from Doppler cloud radar; (c) Attenuated backscattering coefficient from ceilometer. No evidence of clouds in the ABL can be identified by the instruments.

Once the data are curated, the different cloud types are identified. Recent studies employed a cloud classification algorithm by profile with the aim of classify clouds (Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.) and analyze their statistics. However, this algorithm admits different cloud typing inside the same cloud, assigning similar physical properties for different cloud types. In the appendix A, we describe this algorithm, adapting the criteria used by Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.. In this study we propose a new methodology to overcome these inaccuracies. We define a cloud as a group of at least 100 pixels identified by Cloudnet as droplet, drizzle (see Fig. 3). To do so, a cloud mask is generated from the CLOUDNET target classification. Cloud pixels separated only by two pixels in any direction (analogous to 1 min, and 20 - 50 m of height) are considered part of the same cloud. This is achieved using a pixel dilation and contraction technique (Dougherty), which expands and contracts the cloud shapes (see Appendix B for a more detailed explanation of the method). Once the cloud pixels are grouped, each cloud must have more than 100 pixels to be considered (without considering Drizzle or Rain pixels). Otherwise, it is deemed as noise. In order to illustrate this method, the left panel of Figure 3 shows the Cloudnet target classification product (on 11 February 2021), while the right panel shows all groups of hydrometeors identified for this data. Groups inside the black dashed lines are considered clusters and will be classified into cloud types. Additionally, clusters that are very close to each other will be merged to form one cloud. Small and isolated groups of hydrometeors with less than 100 pixels are identified as noise, and are marked by the red dashed lines in Figure 3.b.

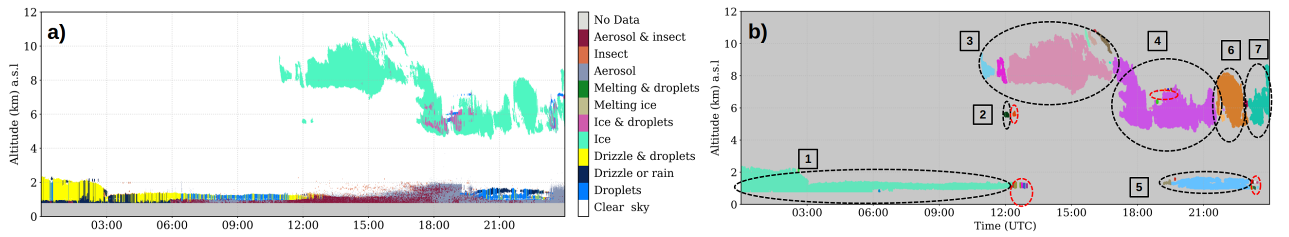

Figure 3. Cloud cluster identification by Cloudnet target classification on 11 February 2021. (a) Target classification, which shows particles typing in time and altitude. (b) Cloud clusters identified in black (dashed lines) and noise (less than 100 pixels connected) by red, dashed lines

Once the clouds are identified, each cloud is classified as one of these six types: liquid, ice, mixed-phase, precipitating liquid, precipitating ice, and precipitating mixed-phase. To do so, the hydrometeor type percentages are calculated for each cloud, (shown in figure 4). A cloud is classified as liquid cloud (Liquid criteria) if it comprises more than 70% of Droplets and Drizzle plus Droplets. Then, it is further classified as a precipitating liquid cloud (Rainy criteria) if it contains at least 10 connected Rain pixels. Similarly, a cloud is classified as an ice cloud (Ice criteria) if the previous verification is not satisfied and if this group contains more than 90% of ice or less than 10% of Droplets plus Ice & droplets. The same rain verification as in the liquid case is made to classify this group as a pure ice cloud or a precipitating ice cloud. Otherwise, clouds are classified as mixed-phase or precipitating mixed-phase through the same rain verification as in the other clouds. In relation to precipitating ice and precipitating mixed-phase clouds, an extensive analysis of individual case studies revealed that Drizzle & droplets only appear below the melting layer together with Drizzle & rain droplets. Therefore, to avoid an underestimation of cloud base height due to drizzle, we removed these pixels for these clouds since they are probably rain.

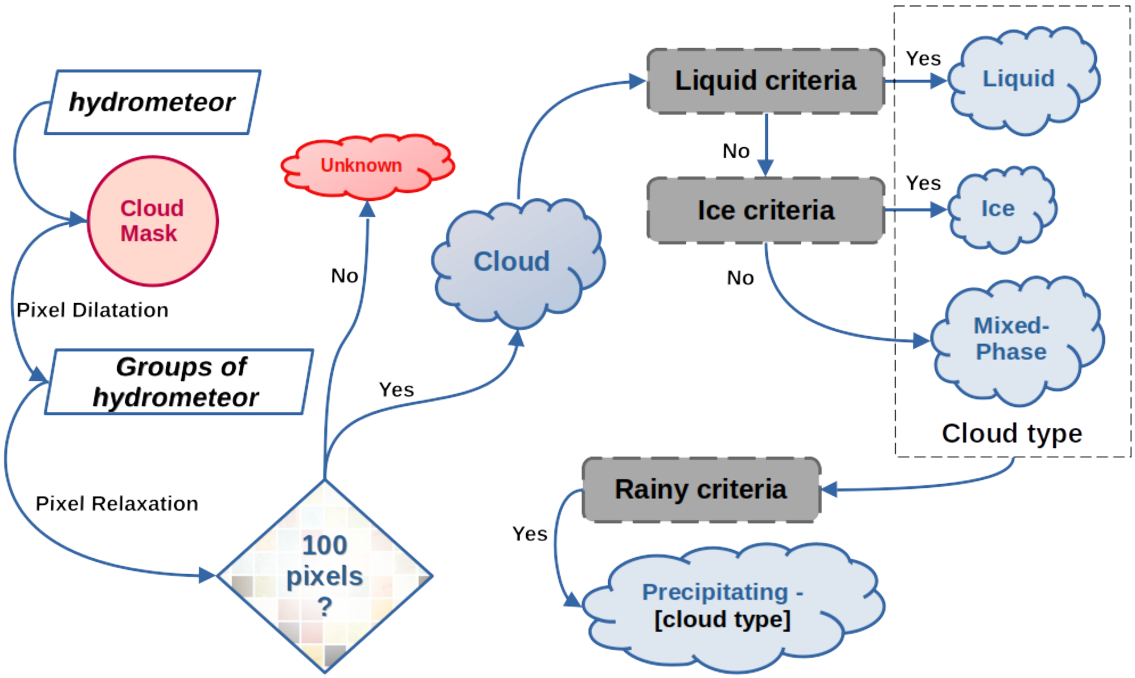

Figure 4. Scheme of the cloud classification algorithm

After the cloud classification, each cloud is then classified as single or multi-layer over time. Initially, time intervals where clouds overlap are identified. Overlapped pieces of clouds are classified as multi-layer, whereas non-overlapping pieces in principle are classified as single layer. If a cloud had a second layer separated by only one pixel at some time interval, that interval is identified as containing multi-layer. Otherwise, the clouds at that interval are classified as a single layer. One pixel separation as threshold was settled to avoid any possibility of considering multi-layers.

In this study, only single layer clouds are analyzed. For the single layer case, cloud properties are calculated as follows: the cloud base and top height are determined by the first and last height pixels within the cloud layer, respectively. Cloud thickness is obtained by the difference between cloud base and top height pixels.

4.2 Cloud typing assessment

An evaluation of the cluster classification algorithm presented in this study against the profile-based method used by previous studies is conducted in order to evaluate its applicability for further analysis. Here, we aim to provide insights into the correlation on cloud statistics between different cloud categories generated by these two methods. In this framework, Figure 5 shows a case study on 6 May 2023 in Granada comparing profile-based cloud classification and cluster classification for a mixed phase cloud, where red and black dots represent the cloud base and top, respectively. As expected, the profile-based classification on the right panel leads to the fragmentation of a single cloud into different types. In this case, the same cloud is being classified as ice and mixed-phase. On the other hand, in the left panel, the new classification method preserves the homogeneity of the cloud and classifies it as mixed-phase cloud, resulting in a more realistic classification.

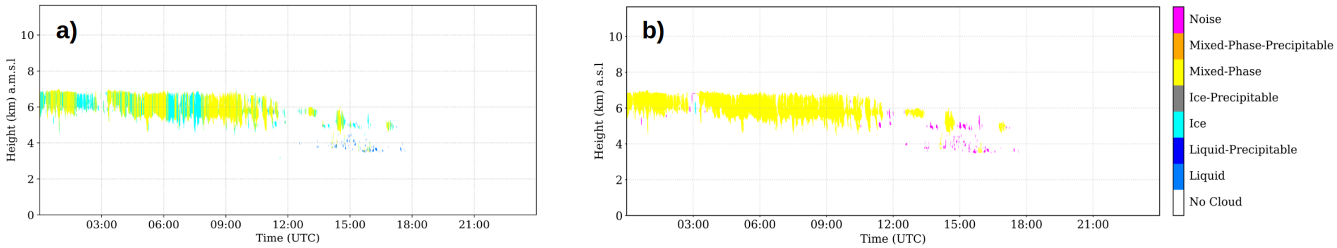

Figure 5. Comparison of cloud classification methods. The left panel illustrates the new method, which preserves cloud homogeneity and provides a more realistic classification. The right panel shows the old method used in recent studies, where cloud classification is time-based.

As a consequence, the profile-based method of cloud classification can generate high correlations between cloud categories in terms of frequency of occurrence and cloud properties, which are not realistic. For example, the right panel of Figure 5 suggests that both ice and mixed phase clouds will have similar daily cloud occurrence and average cloud base. However, this is occurring because of the classification method, which could further affect the analysis of cloud type properties reaching misleading conclusions.

We calculated the daily cloud occurrence for each cloud type to quantify the correlation between cloud categories generated by these two classification algorithms for the whole database in order to assess the cluster-based method proposed in this study. Figure 6 shows the pearson correlation coefficients of daily cloud occurrence between cloud types for the cluster classification on the left, and for the classification by profile on the right. As it can be seen, the cluster method presents generally weaker correlations among cloud types, with most values clustering around zero. For example, the highest correlations observed were 0.19 between Liquid and Liquid-Precipitable clouds and 0.17 between Liquid and Mixed-Phase-Precipitable clouds, indicating relatively weak relationships. Additionally, some negligible negative correlations were observed, e.g. between Ice and Mixed-Phase-Precipitable clouds (-0.1). This suggests that the new method may provide a clearer distinction between cloud types, which is advantageous for analyses requiring precise classification. In contrast, the profile-based classification exhibited stronger and more pronounced correlations. Notably, the correlation between Ice and Mixed-Phase clouds was 0.4, and between Liquid and precipitating liquid clouds was 0.32, highlighting significant interdependencies between these cloud types. These relatively higher correlations are a consequence of the classification scheme defined in the profile-based method, which allows tagging multiple cloud types inside the same cloud. As a result, the comparison indicates that the new method, with its generally weaker correlations, may be better suited for distinguishing different cloud types. These findings underscore the importance of selecting an appropriate analytical method to study cloud properties between different cloud types.

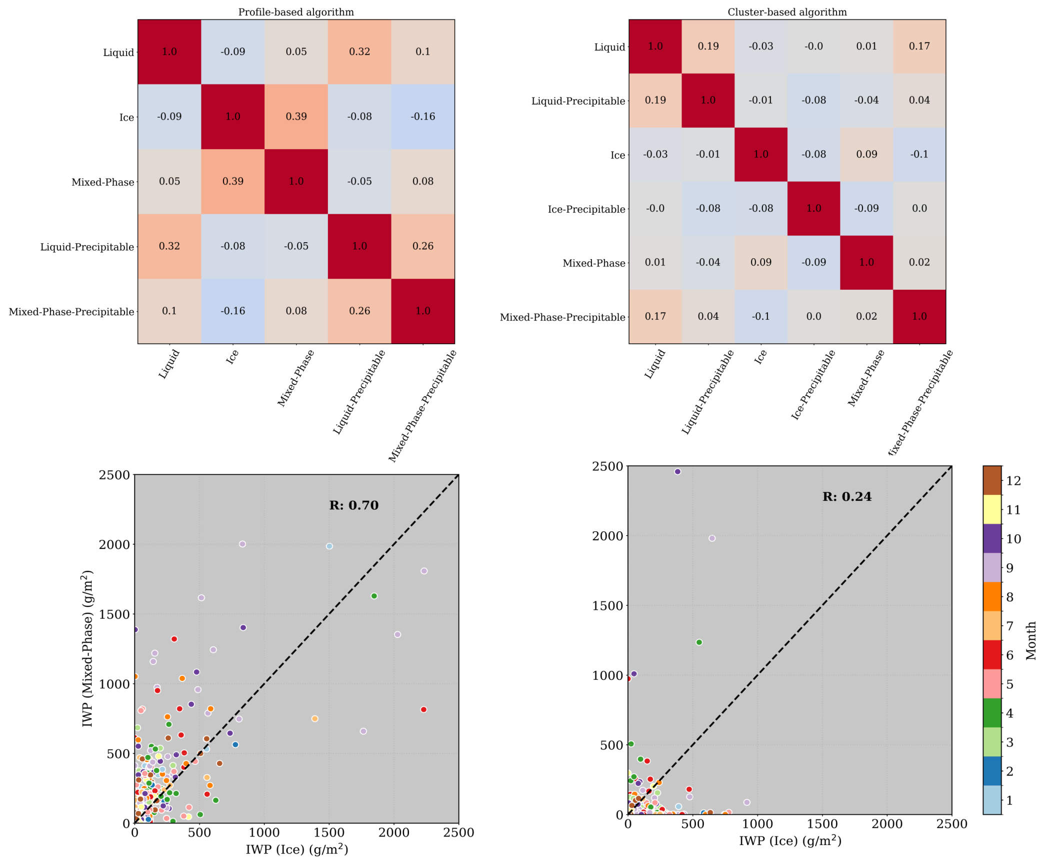

Figure 6. The Pearson correlation matrix of daily cloud occurrence among different cloud types is shown in the top panel. The correlations for cluster-based methods are on the left and the profile-based correlation is on the right. Below each correlation matrix, the corresponding comparison of IWC between ice and mixed phase is shown.

Regarding the correlation in cloud properties, we computed the daily average of IWP for ice and mixed-phase clouds for the two classification methods. The bottom panel of Figure 6 shows the IWP of mixed phase clouds in function of the IWP of ice clouds. The R factor is the Pearson correlation coefficient between the IWP of these two cloud types. This coefficient is quite higher for the profile-based method, showing a correlation of 0.7. On the other hand, the cluster method exhibited only 0.24, with no apparent correlation. In addition, the correlation seems to be not monthly dependent as there is no clear separation by month. It is worthy to note that this section is not focused on the analysis of IWP, this will be done in section 5.3. Instead of this, here we focused on the influence of the classification algorithm on cloud properties analysis. Following, the observations suggest that statistics on cloud properties can be highly influenced by the profile-based classification. As a result, this method may lead to the misinterpretation of real cloud properties, raising the relevance of previous evaluation of cloud classification algorithms before doing statistics on cloud properties. In this light, the new approach shows a promising classification method and will be employed in this study.

5 Results and discussion

5.1 Annual Cloud Variability

Figure 7a presents an overview of cloud occurrence at the AGORA station, showing the monthly occurrence frequency for different sky conditions, i.e. single-layer clouds, multi-layer clouds, clear sky, and noise (as defined in section 4.1). The data reveal a distinct seasonal pattern in cloud cover and clear sky occurrences. Clear sky conditions (green line) are predominant most of the year, being most frequent in the summer months, peaking in July and August (> 80%), with a significant decline in spring, reaching the lowest point in March and April (40%), when cloudy skies are predominant. This pattern indicates a higher prevalence of clear skies during the warmer months and increased cloudiness during the colder months. Single-layer (blue line) and multi-layer clouds (orange line) exhibit very similar values with a relatively weak seasonal pattern throughout the year showing maxima in spring and minima in the summer months, Multi-layer clouds present a slightly more pronounced seasonal variation than single-layer clouds, with higher occurrence in spring (35% for multi-layer and 25% for single-layer clouds in April) and lower occurrence in summer (3% for multi-layer versus 9% for single-layer clouds in July). The noise category (red line) presents the minimum values and maintains a weak seasonal pattern throughout the year. Overall, the data highlight clear seasonal variations in sky conditions, with predominating clear skies throughout the year and increased cloudiness, both single and multi-layer, in spring and winter. Although AGORA exhibits a considerable multi-layer occurrence in spring, these clouds are problematic due to inaccurate LWP partitioning among cloud layers (Shupe et al.). The partitioning needs an accurate cloud profile, which cannot be assessed for elevated clouds with only radar reflectivity. The reflectivity is not sensible to small cloud droplets and the lidar signal is totally attenuated by the first layer. Hence, liquid layers will be poorly identified causing an inaccurate determination of liquid and mixed phase clouds as observed by Tatiana Nomokonova et al.. Therefore, the following analyses are focused only on single-layer clouds, since their physical properties can be retrieved more accurately.

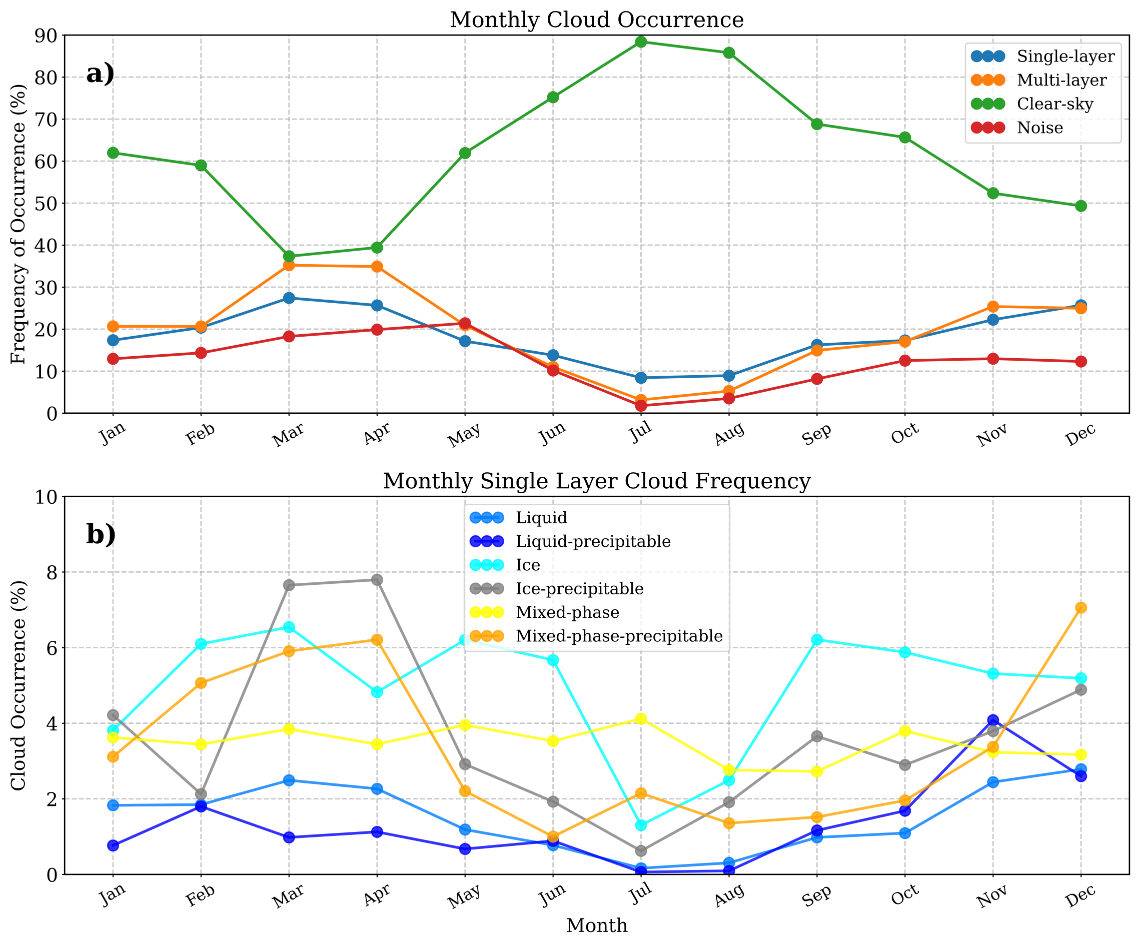

Figure 7. a) inter-annual frequency of occurrence for single layer clouds (blue), multi-layer clouds (orange), clear sky (green) and noise (red). b)inter-annual single layer cloud occurrence for different cloud categories (see section 4.1). Each cloud types is indicated in the legend

Monthly occurrence of single layer cloud types is depicted in Figure 7b. The percentage of occurrence in this figure is calculated in relation to the total number of data. Thus, the sum of cloud occurrences per month must retrieve the blue curve in Figure 7a., which is the single layer frequency in relation to the total number of data. Mixed-phase-precipitable clouds (orange) peak in April (6%) and December (7%), indicating higher chances of mixed-phase precipitation towards spring and the end of the year. Meanwhile, mixed-phase clouds (yellow) exhibit a relatively consistent occurrence with slight fluctuations around 3% and 4%. Liquid-precipitable (dark blue) clouds have a modest variation with zero occurrence in summer, followed by an increase during the transition to winter, reaching a maximum of 4% in November, suggesting increased precipitation during fall. Liquid clouds (blue) have quite a similar percentage, with 3% in March and December, against zero in summer. Ice-precipitable clouds (gray) have a notable peak of 8% in spring, followed by a decrease in summer (1%) and an increase in fall, until they reach a local maximum of 5% during winter. It suggests seasonal variations in ice-related precipitation with maximum in spring, unlike the case of water-related precipitation, which showed a maximum in November. Ice clouds (cyan) drop sharply in July and August and maintain almost constant levels around 5% and 7% throughout the rest of the year, indicating that hotter temperatures inhibit ice cloud formation during summer. Overall, ice, precipitating ice, and precipitating mixed-phase clouds have the strongest seasonal pattern, with a predominance of these types of clouds over liquid clouds throughout the year. Moreover, the precipitation in this region is intrinsically related to ice and mixed phase clouds, occurring mostly in spring. During summer, non-precipitating clouds are observed. Together with the elevated summer temperatures, the Iberian Peninsula experiences a large low-pressure area over northern Africa, with high-pressure systems over the Atlantic Ocean and eastern and central Europe that results in easterly winds over Andalusia, transporting moisture from the Mediterranean Sea in summer (F. J. Bello-Millán et al.). This combination of high temperatures and humidity can create an environment conducive to storm formation in Granada during the summer. In this context, the few clouds observed in summer are likely related to convectively formed cumulus or cumulonimbus clouds.

The analysis of the diurnal cycles revealed no clear diurnal patterns, thus no analysis on diurnal cycle is shown. Overall, liquid and liquid-precipitable clouds are the least likely, independently of the daily hour. On the other hand, Spring shows a slight increase in Ice-precipitable and Mixed-phase-precipitable cloud occurrence during daylight hours. However, this increase was not significant enough to analyze the diurnal cycle.

On the other hand, other single layer clouds have less than 4% of occurrence at this time. In spring, ice, mixed-phase and ice-precipitable clouds present a higher occurrence. Spring shows a slight peak of occurrence for Ice-precipitable and Mixed-phase-precipitable clouds between 09:00 and 21:00 UTC. Different from non-precipitating clouds which reached a minimum around mid-day, these clouds doubled the occurrence from at 00:00 (4%) to 16:00-17:00 UTC (8%). Despite that these observations presented small diurnal variations, they are consistent with the findings of Shahi et al., where convection-permitting models of the Iberian Peninsula (IP) captured high values of mean precipitation during spring. Additionally, Alexandre Rios-Entenza et al. revealed that specific synoptic conditions, such as a reduction in the strength of the zonal circulation, can favor local and mesoscale convection during spring in the western region of the IP. This analysis confirms that the maximum cloud and rain occurrence observed during peak sunlight hours may be strongly associated with local convection.

5.2 Seasonal analysis of cloud microphysical properties

Seasonal evolution of cloud base height, top height and thickness are displayed in Figure 8. Each color represents a cloud type, where liquid clouds are in blue, ice clouds are in gray and mixed-phase clouds are in orange. Also, the lightest colors represent non precipitating clouds, and the darkest colors represent precipitation clouds. The cloud base analysis (Figs. 8.a, d and g) reveals that precipitating clouds exhibit a significantly lower cloud base height (top panel) compared to non-precipitating clouds, as well as a narrower distribution. Moreover, these precipitating clouds exhibited the most distinct seasonal pattern. Regarding non-precipitating ice clouds, they possess the highest cloud base, with median values around 7 (winter and spring) and 8 km (summer and fall). The ice clouds with base height above the 50/75th percentile can be normally associated with cirrus clouds. Next, non-precipitating mixed-phase clouds have the second highest cloud base, showing considerable variability, except during summer, when values stabilize at 5.5 km. Also, these clouds present a higher base in fall, where the 50th percentile reaches 6.5 km. The precipitating mixed phase clouds have much less variability and higher cloud base at 4.5 km in summer, as expected. For non precipitating liquid clouds, cloud bases (Fig. 8.a) are highly variable during summer, achieving significantly different heights compared with other seasons. Overall, in summer, precipitating mixed-phase clouds, and ice precipitating clouds demonstrate elevated cloud base heights, attributed to the relatively high temperatures and low humidity at this season. Liquid clouds also have elevated base height in summer, however only a few cases were registered, with 50th and 75th percentiles around 4 km and 6 km, respectively. These clouds can be related to stratocumulus and stratus clouds, which were decoupled from cumulonimbus at the final stage in summer.

In general, precipitating clouds also have significantly lower median cloud top height (Figs. 8.b, e and h) compared to non-precipitating clouds and exhibit a narrower cloud top distribution, except in the case of ice clouds. Non-precipitating ice clouds have median cloud tops height fluctuating around 9.5 km through the seasons, while ice-precipitable clouds demonstrate highly variable cloud tops. Given their well-defined cloud bases, these clouds exhibit greater cloud thickness. The cloud tops for liquid clouds mirror the variability of their cloud bases, especially during summer. These clouds exhibit a well-defined cloud thickness (Fig. 8.c – zoomed for better visualization), with non precipitating liquid clouds having a thickness around 150 m and precipitating liquid clouds reaching up to 500 m. Non precipitating mixed-phase cloud tops also present high variability. Nevertheless, they displayed a well-defined cloud thickness, with similar distribution along seasons. Their median cloud thickness is around 400 m, increasing to 500 m for the precipitating case (dark orange), with slightly higher values in summer. In general, non-precipitating mixed-phase clouds have median cloud tops around 7 km during spring, summer, and winter, but reach 8.5 km in fall. Then, precipitating mixed-phase clouds have cloud tops around 3 km in spring, fall, and winter, peaking at 7 km in summer.

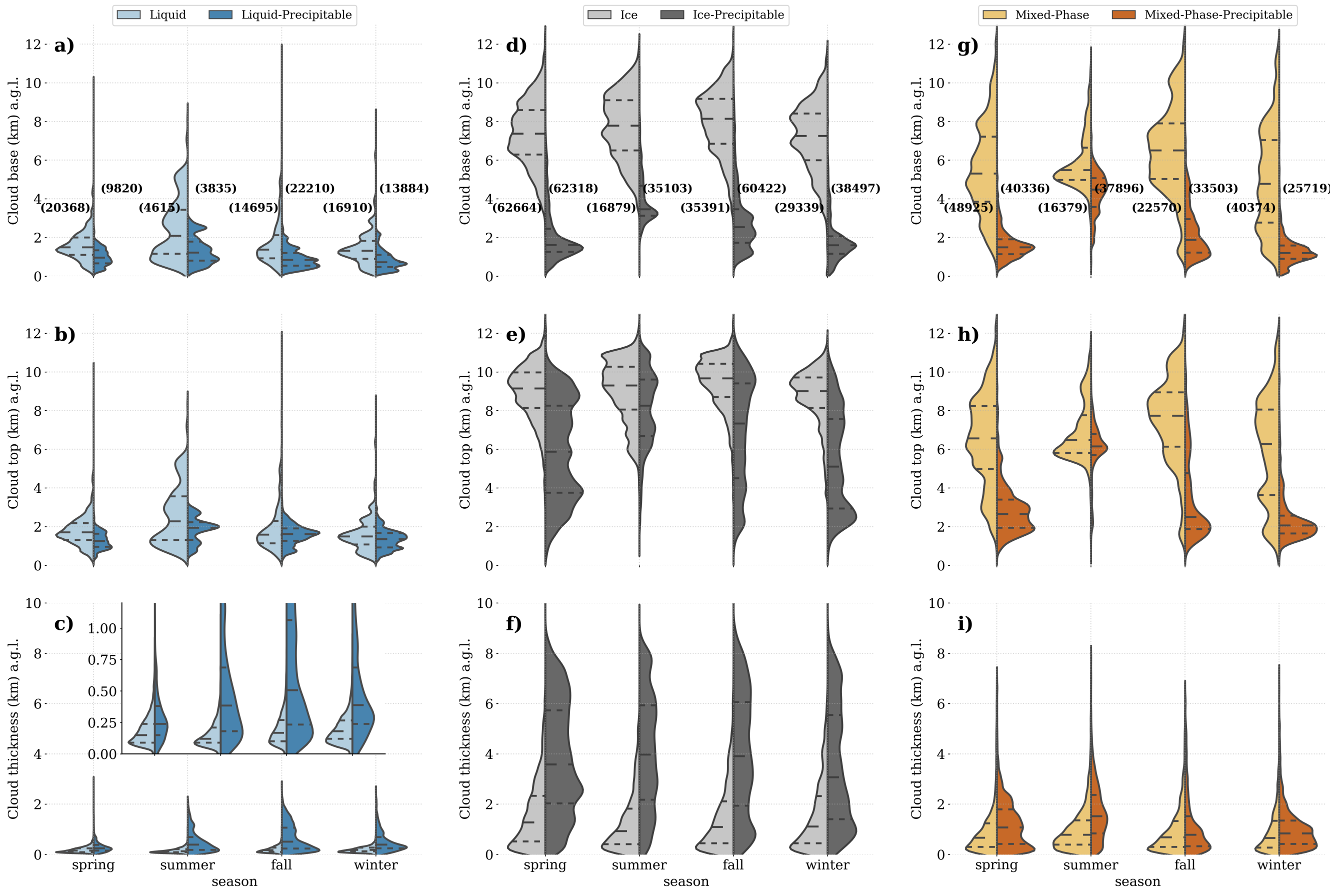

Figure 8. Seasonal evaluation of cloud base height (top), top height (middle) and thickness (bottom). For different cloud types: liquid (left), ice (middle) and mixed-phase (right)Dashed lines indicates 25/50/75th percentiles

Table 3 summarizes the median cloud base heights and thickness found in figure 8. This table shows that ice-precipitable clouds exhibit the largest cloud thickness, followed by Ice, mixed-phase-precipitable, Mixed-phase, Liquid-precipitable, and Liquid clouds. Ice clouds present the highest cloud base height, followed by Mixed-phase, with liquid-precipitable clouds having the lowest base altitude. In general, it is clear that precipitable clouds present much lower base height than their non-precipitable homonyms. It is important to note that, despite Cloudnet post-processed data being corrected for liquid attenuation, cloud tops for precipitating clouds can be underestimated due to high liquid attenuation in some particular cases. In addition, in precipitating liquid clouds, the cloud base is sometimes determined by "Drizzle & Droplets" pixels, which may represent precipitation. It is hard to know whether these pixels are falling or not when they are inside a “Drizzle or rain” layer. In such cases, the cloud base could be underestimated. At the moment, recent studies are trying to overcome this problem, which can be more intense on marine sites (Roschke et al.), but it is true that this is rarely observed at our station. In contrast, this is most common for precipitating ice and mixed-phase clouds, where we do not use these pixels to define the cloud base. Instead, we rely on those within the melting layer, as outlined in the methods section.

Table 3. Cloud macrophysics per season and cloud type. Median values of cloud base height, cloud top height, and cloud thickness are shown.

		Seasons			
Cloud Properties	**Cloud Type**	**Spring**	**Summer**	**Fall**	**Winter**
Cloud base height (km)	Liquid	1.49	2.09	1.37	1.31
	Liquid-Precipitable	0.95	1.22	0.83	0.72
	Ice	7.36	7.78	8.14	7.24
	Ice-Precipitable	1.61	3.46	2.53	1.60
	Mixed-Phase	5.31	5.49	6.50	4.77
	Mixed-Phase-Precipitable	1.49	4.47	1.86	1.19
Cloud thickness (km)	Liquid	0.15	0.12	0.17	0.18
	Liquid-Precipitable	0.24	0.38	0.51	0.39
	Ice	1.27	0.92	1.09	1.10
	Ice-Precipitable	3.58	3.96	3.91	3.07
	Mixed-Phase	0.66	0.78	0.68	0.66
	Mixed-Phase-Precipitable	1.07	1.52	0.78	0.83

The seasonal pattern for cloud base and thickness can be easily observed in table 3. Normally, the cloud types discussed have higher cloud bases in summer, except for Ice and Mixed-phase clouds. Especially for Mixed-phase clouds, they deviate from the expected pattern of having a lower cloud base during fall and winter, when temperatures are lower. Following, Liquid and precipitating Liquid clouds are thicker in fall and winter, while precipitating Ice, Mixed-phase, and precipitating Mixed-phase clouds are thicker in summer. Lastly, Ice clouds are thicker in spring since, showing different seasonal patterns in comparison to Mixed-phase clouds. This behavior differs from the findings of Răzvan Pîrloagă et al., who reported similar values. The discrepancy may be attributed to the method used by the author, as clouds were classified using a profile-based algorithm. We have previously demonstrated that this approach can lead to high correlation in cloud properties between these two cloud types. On the other hand, precipitating Ice clouds do have a similar seasonal pattern to Mixed phase and precipitating Mixed-phase clouds. We have some degree of confidence that this result does not depend on the method, as we have previously demonstrated that there is no correlation between the daily occurrence of these cloud types when using the cluster-based classification method.

5.3 Cloud microphysical properties

In this section we investigate the inter-annual variability of microphysical retrievals for each category of single-layer clouds. The assessed cloud properties are LWC and r_{liq} for liquid and mixed phase clouds, IWC and r_{ice} for ice and mixed phase clouds (of precipitating and non precipitating type).

5.3.1 Liquid clouds

The monthly median profile of LWC shows high variability of LWC in function of height for liquid clouds (Fig. 9.a). LWC is generally around 50 mg m^{-3} during the whole year, with no seasonal pattern. Maximum median values are observed during summer, fall and winter, whereas for spring a maximum is observed between 4 and 5 km. A sharp peak of about 500 mg m^{-3} can be also clearly identified in spring below 500 m. This peak is related to liquid clouds observed during the springs of 2020, 2021 and 2022. For particular months such as February, March and April, quite large values below 1 km and within 2.5 and 4 km (right panel of Fig. 9.a) are observed. Nevertheless, above 500 m, really high values (> 1000 mg m^{-3}) observed in the monthly profiles are not representative of the entire season as can be seen in the left panel of Figure 9.a, where the seasonal median are much lower than monthly peaks. Summer months exhibited a lack of liquid phase clouds, with some elevated LWC at higher levels as shown in the left panel of Figure 9.a. The average liquid water path (LWP) (bottom panel of Fig. 9.a, black star) shows a considerable variability of LWP. Although the median stays stable, the mean is quite variable and different from the median, indicating high variability. It highlights a very wide variety of liquid clouds in these months. A notable increase in the median LWP occurs in April, aligning with their maximum cloud occurrence. From August to October, there is a gradual increase in average LWP, while the median does not change. This trend corresponds with the increased occurrence of liquid clouds from summer to fall, impacted by different mechanisms of cloud formation. Since these months reported just a few occurrences of liquid clouds, as can be checked by the blue line, which shows the number of cloud profiles per month, their statistics will be much more sensible to these mechanisms. As a result, LWP in summer is the smallest, and there is no significant difference between the average and the median.

Precipitating liquid clouds exhibit distinct seasonal vertical profiles of LWC. The left panel of Figure 9.c shows that LWC can reach more elevated heights during spring and fall. In general, this parameter increases with altitude up to 2 km, with the exception of spring, which maintains relatively constant values within this layer. At the top of this layer, LWC is typically 250 mg m^{-2}, and approximately 500 meters higher (2.5 km) during the fall. Above this level, LWC is highly variable with altitude in spring, summer, and fall. In particular, summer demonstrates a second maxima of approximately 250 mg m^{-3} close to 3 km, and a sharp peak near the surface. This is also observed during spring due to the presence of drizzle particles below cloud base during rainfall. It has been attributed to relatively humid months in April of 2020, 2021, and 2022 as already observed for non precipitating liquid clouds. The monthly evaluation on the right-top panel of Figure 9.c supports this idea, in addition to attributing June ($\sim$300 m) as the month causing the sharp peak in summer close to the surface. For the rest of summer and early fall, atmospheric water content is almost zero. For fall, the peaks observed in 2.5 km occur in October, whereas winter peaks at higher levels occur in Nov-Dec and at lower levels in January. Then, in the rest of summer and early fall, atmospheric water content is almost zero. In terms of LWP, the bottom panel of Figure 9.c indicates substantial variability of LWP in January, suggesting a diverse variety of liquid clouds during this month. In contrast, spring exhibits relatively consistent LWP across its months, reflecting stable months for precipitation. In these months, their median stays below 50 g m^{-2}, and summer has even less LWP, except for June, as previously observed.

In terms of droplet effective radius, liquid clouds are well consistent throughout the year, showing small differences in terms of median values (top panel of Fig. 9.b), mostly ranging between 5 and 8 μm. There is no discernible pattern in r_{liq} across seasons, with larger droplet sizes at higher altitudes. We observed a sharp increase of droplets above the cloud base, followed by a slight increase in droplets. In addition, there are almost no liquid droplets above 4 km, with just a few occurrences in summer and early fall registered. The median r_{liq} is generally stable around 5 μm, with a small decrease in July, when almost no clouds were registered. The seasonal profiles confirm the uniformity in r_{liq} distribution with height, showing minimal variation with altitude. It suggests a consistent cloud composition and droplet size throughout different atmospheric conditions. These findings underscore the stable nature of liquid clouds in terms of droplet size and LWC. Normally, these clouds have a small thickness, as observed in Figure 8.b, associated with a small range of r_{liq}.

Different for the purely liquid clouds, precipitating liquid clouds demonstrated different seasonal profiles of median r_{liq} (see left panel of Fig. 9.d). Below 1km, spring and summer have similar r_{liq}, fluctuating around 5 μm. Above this level, droplet radius increases with highest rate in summers up to 2 km, while spring keeps r_{liq} roughly around 5 μm until 3km, followed by sharp increase to 20 μm. Winter has the second largest r_{liq} at the first 2 km ($\sim$8 μm), disregarding large drizzle particles close to the ground. These particles cause sharp peaks near the surface in summer, fall and winter. In addition, r_{liq} in fall are the largest (below 3 km), reaching 15 μm around 1 km. The monthly profiles in the right panel of Figure 9.d shows that larger values in this season occur in November and close to the ground. These sizes are normally related to drizzle droplets as already observed. Their monthly size distribution in the bottom panel of Figure 9.d displays different size distributions over the year, where higher median values are found in fall as already observed, specially in November. During spring and early summer, the median is maintained around 10 μm, then decreases in summer, and increases again in fall, when the occurrence of this cloud type increases, while temperatures decrease.

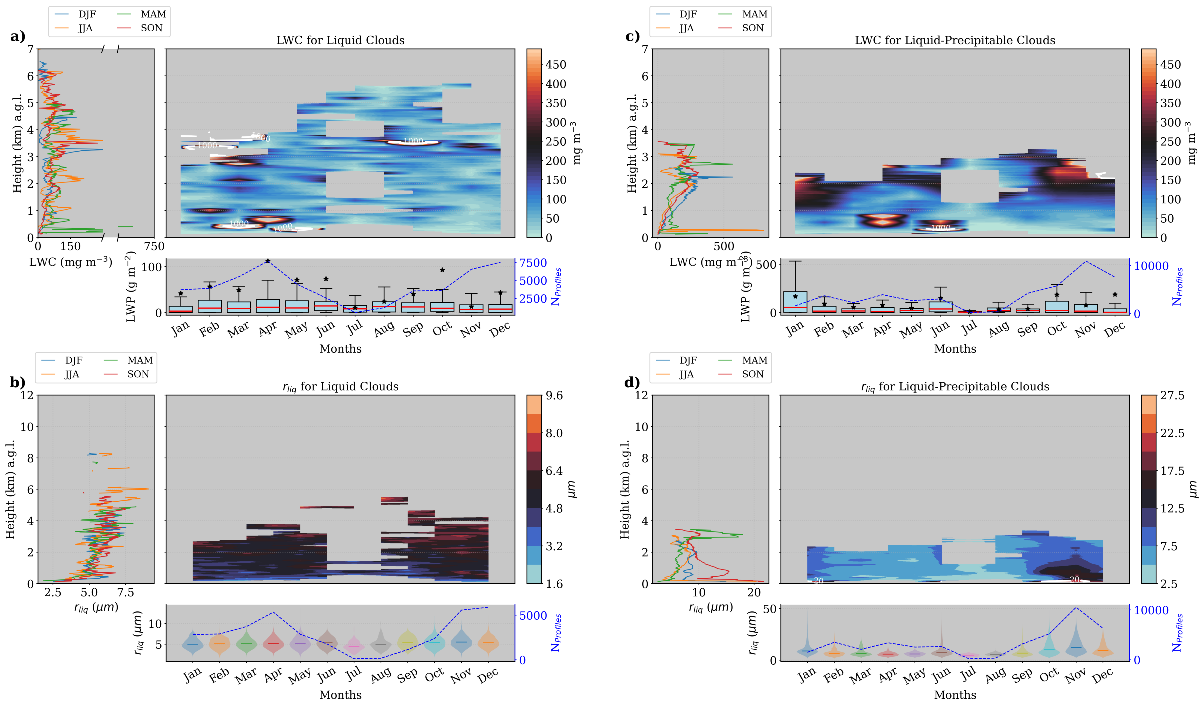

Figure 9. Seasonal and monthly average profiles of microphysical properties, along with monthly of LWP and r_{liq}. Panels a) and b) show LWC, r_{liq}, and LWP for non-precipitating liquid clouds, while panels c) and d) depict the same parameters for precipitating liquid clouds. Specifically: a) and c) display the seasonal (left panel) and monthly (right panel) median profiles of LWC, with boxplots of LWP beneath. Medians are represented by red lines and means by black stars. b) and d) present profiles of r_{liq} for liquid and precipitating liquid clouds, respectively. The left panels show seasonal medians, while the right panels show monthly medians. The monthly distribution of r_{liq} is illustrated using violin plots in the bottom panel.

Overall, Liquid clouds are challenging to analyze due to their low frequency of occurrence. They exhibit significant variability with height, making it difficult to discern seasonal differences in LWC profiles, except at low levels. Near the surface, LWC is much higher in spring, and their droplet size (r_{liq}) remains relatively constant around 5 μm above the cloud base. In contrast, precipitating liquid clouds display a more pronounced seasonal pattern, with higher LWC and larger r_{liq}. Notably, the presence of drizzle particles near the surface makes r_{liq} larger. During summer, these clouds exhibit two distinct peaks in LWC in function of height, one at 2 km and another at 3 km

5.3.2 Mixed Phase clouds

The LWC for Mixed-phase clouds have large variability below 4 km and above 10 km (see left panel of Figure 10.a. Particularly during spring and summer, this variability is significantly greater than in other seasons, as evidenced by the two pronounced peaks near the surface. These peaks are located around 500 m, and reach values of 500 and 1000 mg m^{-3} (even greater than the maximums found for liquid clouds). Next, the monthly average profiles on the right panel also show extremely high values of LWC at different levels. Despite the fact that these values are more frequent for mixed-phase clouds than liquid clouds, the LWP (bottom panel of Fig. 10.a) of both is quite comparable, being slightly larger for liquid clouds. Since mixed-phase clouds have a nearly constant occurrence rate of around 3-4%, this suggests that their variability in liquid water content (LWC) is not linked to monthly frequency, but rather to their inherent physical properties. This variability can also be checked by the monthly difference between the mean (black star) and the median (red line) LWP in the bottom panel.

Regarding the ice water content (IWC), it exhibits much more variation with height than LWC (see figure 10.c). It increases with altitude, starting at 2 km and growing up to 7 km for colder months (winter and spring) and up to 8 km for warmer months (fall and summer). Winter and fall follow practically the same comportment, as well summer follows the same behavior of fall up to almost 8 km. Above this level, summer has the largest amount of IWC, with a peak of 15 mg m^{-3} around 11 km. However, ice mass is generally greater at middle and high altitudes. The right panel of Figure 10.c shows significant values of IWC of about 20-40 mg m^{-3} between 4 and 10 km, except for July. As for the LWC, July also presented elevated IWC above 10 km (more than 60 mg m^{-3}). These clouds might be related to cumulus clouds, taking into account they have considerable amounts of ice, especially at their top. In addition, these clouds have a considerable thickness, some of them reaching 2 km of thickness. Furthermore, the bottom panel of Figure 10.c shows that median values of IWP are comparable to the LWP. However, they are much less variable, which can be seen by the smaller differences between the median and the mean monthly IWP. In general, typical values are below 10 g m^{-2}, being slightly larger in April and summer, when there are more observations.

The r_{liq} of Mixed-phase clouds (Fig. 10.b, left) increased with height up to 5 km, where there is almost zero LWC as already observed. This profile highlights a general behavior that can be explained as follows: As temperature decreases with height, ice particles start to form, increasing competition for water vapor and halting the growth of liquid droplets to form ice particles as temperature continues to decrease. It will decrease the size of water droplets keeping their LWC fixed (close to zero between 4 and 10 km) while the ice mass concentration increases (see left panel of Fig. 10.c). The right panel of Figure 10.b allows a more detailed evaluation of r_{liq} per month. Large droplet sizes are located between 4 and 8 km, except for July, which also shows larger droplets at 10 km. As discussed, liquid water is uncommon at these levels, suggesting the presence of clouds with large vertical development as cumulus clouds. Overall, summer exhibits larger droplets compared to colder months, with an effective radius between 14-22 μm at medium heights (> 4 km and < 6 km), and between 12-18 μm around 10 km. Conversely, droplet sizes are generally close to 10 μm, as is shown by the monthly distribution in bottom panel of Figure 10.b, where the modes are at even smaller values than the median. In fact, cloud droplets are slightly larger in summer, with a good probability of reaching 15 μm, but generally, they present smaller droplets.

In terms of r_{ice}, their size follows a different function of the height. The seasonal profiles (on the left panel of Fig. 10) show a smooth decrease of effective radius with height up to 8 km, and a faster decrease above this layer. In the entire column, summer has a slightly larger radius, especially at 11 km, reaching 30 μm, whereas other seasons showed smaller radius at this level. The monthly analysis (see Fig. 10) undoubtedly shows that higher radii are at low levels, except in July where it's possible to find r_{ice} larger than 50 μm above 10 km. However, the gradient of colors shows that normally, ice clouds have larger ice particles the closer to the surface, where there is low amounts of ice water content (IWC). In addition, differently from water droplets, which have shown monthly distributions of r_{liq} with the median above the mode, r_{ice} shows a median below the mode (see bottom panel of Fig. 10). Overall, the median is fluctuating around 45 μm, with the mode above this value, indicating high probability to find an elevated radius.

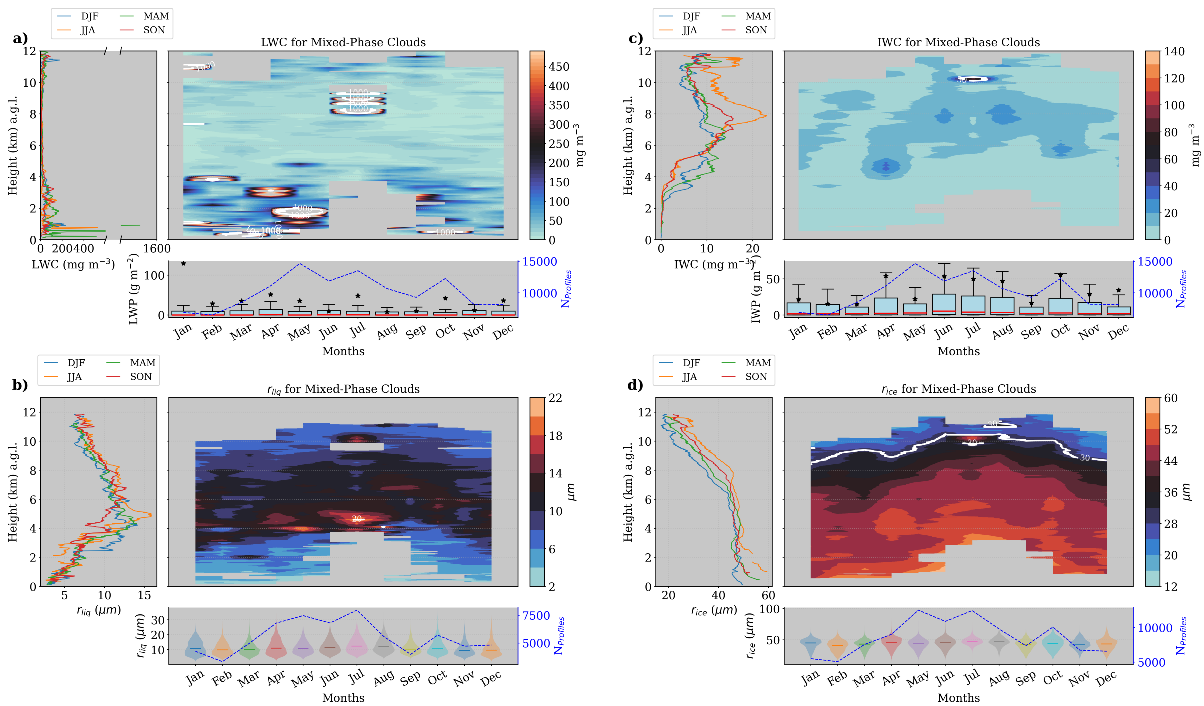

Figure 10. Same as Fig. **??**, but for mixed-phase clouds.Seasonal and monthly average of ice water content (IWC) and ice water path (IWP) for mixed-phase clouds. Left panel shows the seasonal median profiles of IWC. Right panel shows the monthly median profiles of IWC and monthly box-plot of IWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

Similarly, the seasonal and monthly evaluation of liquid and ice properties of precipitating mixed-phase clouds is shown in Figure 11.a. In the left panel of this figure, spring exhibited the largest values of LWC, with a sharp increase that surpasses 1000 mg m^{-3} near the surface. Additionally, a secondary, much less pronounced peak is observed at 2 km during this season. At low levels, fall also exhibited elevated LWC values, with a median around 300 mg m^{-3} at 500 m. During this season and also summer, considerable values of LWC are found just above 4 km, whereas winter and spring have LWC close to zero just below 4 km. Thus, warmer and dry months (such as summer and fall) cause liquid particles to form at higher levels. As previously mentioned, the temperature in Granada during summer strongly increases, making it difficult for clouds to form, as they have to reach higher altitudes. Additionally, summer reveals a clear water shortage, especially at low levels. Both in summer and fall, considerable amounts of LWC are present at medium levels in the atmosphere (> 2 and < 4.5 km). This can be easily seen in the left panel of Figure 11.a, where fall (in red) and summer (orange) have almost the same vertical profile, varying around 200 mg m^{-3} between 2 and 4.5 km. The monthly profiles in the right panel of Figure 11.a displays that precipitating mixed-phase clouds generally have LWC at lower levels than non-precipitating mixed-phase clouds (practically zero below

5 km). Additionally, they present a stronger seasonal pattern and larger LWC, which is natural given their strong seasonal pattern in frequency of occurrence. It is worth mentioning that the color scale in left panel of Figure 11.a is the same as in the non-precipitating case in Figure 11.a, right panel. Moreover, these precipitating clouds do not present such high variability with height, consistent with their cloud base distribution, showing much less variability than the non-precipitating clouds. The integrated liquid water path (see bottom panel of Fig. 11.a) exhibits high variability in colder months, with medians below 50 mg m^{-2} and averages above 100 mg m^{-3}. In summer, just a few occurrences of these clouds were observed, resulting in less variability, with both medians and averages close to zero.

In terms of r_{liq}, seasonal profiles are quite different. Large size droplets are observed close to the surface on account of drizzle particles below the cloud base. Since these clouds usually have the base height above 2km, their r_{liq} is about 8 μm (see left panel of Fig. 11.b). Below this level the liquid droplets are larger during summer, and practically the same among the rest of the seasons. Above this level, summer and fall have the largest droplets (> 4 km and <6 km), with maximum sizes at close to 4.5 km. Spring also has the maximum close to this layer, although their radii size is smaller and comparable with the winter maximum at 4 km. Around this layer and also shortly above, the left panel of Figure 11.b indicates larger droplets in January, February and April. For summer, larger values of r_{liq} can be clearly identified around 2 km in August (30 μm) and at 4.5 km in July (20 μm). For spring, droplet sizes are larger close to 5 km, with values slightly smaller than 20 μm. Bear in mind that we are not considering droplets sizes below 2 km, since they are probably related to rain pixels, but classified as drizzle droplets by Cloudnet. In general, the monthly distribution of r_{liq} shows a median around 10 μm, reaching 15 μm in July, August and September.

Following, the left panel of Figure 11.c reveals that the ice amount inside these clouds is found all above 3 km, and exhibit different seasonal profiles. Typically, IWC profiles for summer and spring are similar up to an altitude of 9 km. Just above this level, summer profiles exhibit a distinct peak, exceeding 200 mg m^{-3} of ice. Below this level, spring and winter have similar profiles with less amount of ice. These seasons have a maximum of 22 (winter) and 33 mg m^{-3} (spring) at 4km, respectively. In addition, it is possible to observe a large layer during colder months (winter-spring) with small values fluctuating around 15 mg m^{-3} from 5.5 to 7.5 km. On the other hand, summer and fall presented a maximum of 40 mg m^{-3} at 7 km and 50 mg m^{-3} at 5 km, respectively. Above 8 km, the peak observed in summer is related with the highest values of IWC in June as can be seen in the right panel of Figure 11.c. However, this month has the smaller data availability (see the blue line in the bottom panel), indicating less statistical confidence. Since the entire summer has small data availability, this maximum is reproduced in the vertical profile on the left panel. In fall, the highest values of IWC occur in November, just above 5 km, and ranging from 60 to 100 mg m^{-3}. For winter, the maximum IWC occurs at 5 km in January, reaching more than 120 mg m^{-3}. However, since we cannot observe this peak on the winter profile (on the left panel), January represents just a particular case rather than typical winter conditions. Particularly, July, August, and September have the most comparable largest amounts of IWC in the atmosphere. Consequently, the 25th, 50th, and 75th percentiles during these months are relatively higher, with 25% of the data showing IWC values exceeding 200 mg m^{-2}.

As for the IWC, the r_{ice} have a clear seasonal pattern. It suggests that in general, the ice content has a stronger seasonal pattern than liquid content. The left panel of Figure 11.d shows nearly constant r_{ice} around 50 μm up to 8 km. Above this level, for the warm months, particle sizes decrease in fall while exhibit a sharp maximum at 9.5 km (60 μm) in summer, coinciding with the maximum of IWC. In contrast, winter and spring have similar r_{ice} values to those in summer and fall up to 4 km, after which the sizes decrease with height, reaching 20 μm at 10 km. It is also clear by monthly evaluation that r_{ice} decreases with height, as can be seen on the right panel of Figure 11.d. Particularly during summer and fall, particle sizes exhibit less variability with height, ranging around 50-55 μm. Conversely, winter and spring show variability between 45 and 55 μm. Despite this variability of r_{ice} with height, the monthly distribution shows that within these clouds, r_{ice} is generally around 55 μm.

As for the IWC, the r_{ice} has a clear seasonal pattern. It suggests that, in general, the ice content has a stronger seasonal pattern than liquid content. The left panel of Figure 11.d shows nearly constant r_{ice} around 50 μm up to 8 km. Above this level, during the warm months, particle sizes decrease in fall while exhibiting a sharp maximum at 9.5 km (60 μm) in summer, coinciding with the maximum of IWC. In contrast, winter and spring have similar r_{ice} values to those in summer and fall up to 4 km, after which the sizes decrease with height, reaching 20 μm at 10 km. It is also clear by monthly evaluation that r_{ice} decreases with height, as can be seen on the right panel of Figure 11.d. Particularly during summer and fall, particle sizes exhibit less variability with height, ranging around 50-55 μm. Conversely, winter and spring show variability between 45 and 55 μm. Despite this variability of r_{ice} with height, the monthly distribution shows that within these clouds, r_{ice} is generally around 55 μm.

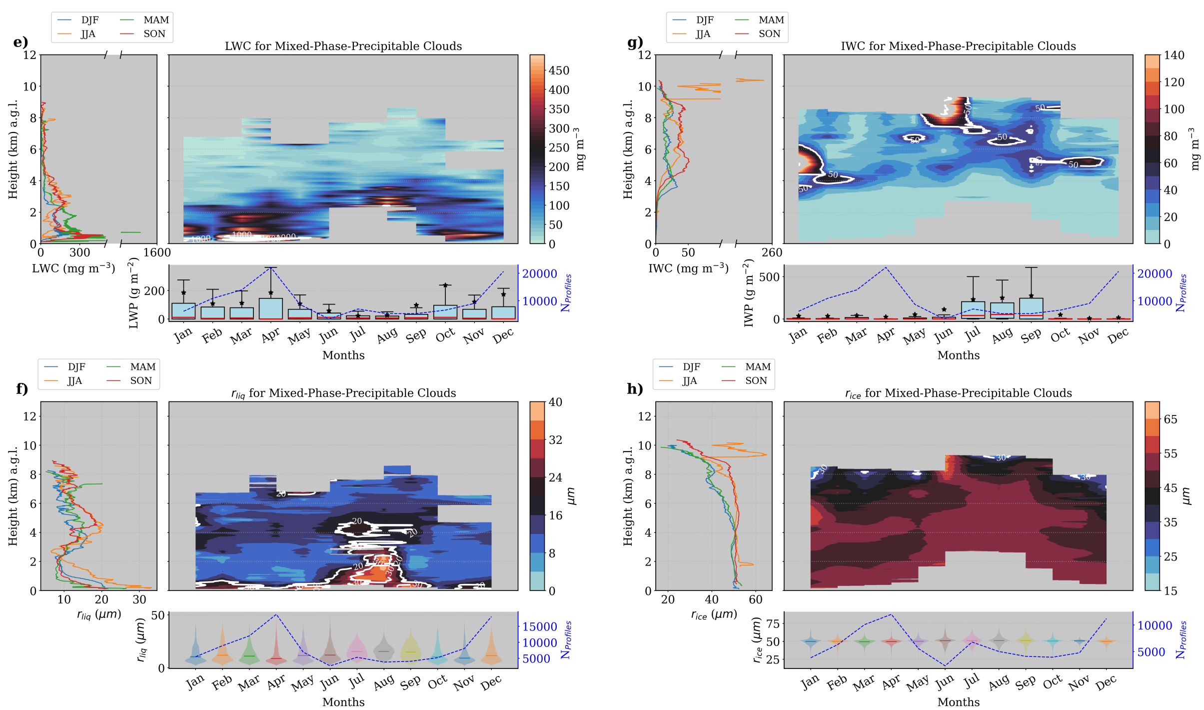

Figure 11. Same as Figure 10, but for precipitating mixed-phase clouds.

Overall, Mixed-phase clouds exhibit less liquid water content (LWC) than liquid clouds. Typically, the LWC for mixed phase clouds is concentrated at low levels, with some exceptions in July, where high LWC values are observed above 8 km. The droplet effective radius in mixed-phase clouds is larger at mid-levels, while the ice effective radius is only large in mid-levels during winter and spring, and larger at 8 km during fall and summer. Unlike this pattern, the ice effective radius in these clouds slightly decreases with height up to 6 km, where the rate of decrease intensifies. Precipitating mixed-phase clouds exhibit a stronger seasonal profile, with higher LWC, particularly in spring and fall at lower levels due to the drizzle presence. They also show significantly more IWC in summer and early fall. The droplet effective radius (r_{liq}) in precipitating mixed-phase clouds is typically higher than in non precipitating mixed-phase clouds, especially near the surface and at mid-levels, due to the presence of drizzle particles at low levels. In terms of the ice effective radius (r_{ice}), these clouds display a clear seasonal variation, with higher r_{ice} values in summer and fall above 4 km. Additionally, precipitating clouds show larger IWC and r_{ice} values at 9 km in summer, suggesting a possible association with cumulus clouds.

5.3.3 Ice Clouds

Regarding ice clouds, high amounts of ice during winter create a strong seasonal pattern of IWC for ice clouds. In the left panel of Figure 12.a, a peak of almost 500 mg m^{-3} of IWC is observed at 1 km. Despite the fact that non-precipitating ice clouds are usually related to cirrus clouds (located at high levels), winter shows low-level ice clouds with much more IWC. However, they represent a minor portion of ice clouds, since we already observed that only 25% of them have base heights below 6 km. Above 3 km, the profiles of winter and spring are quite similar, having a maximum of only 25 mg m^{-3} of IWC at 6 km. Fall has two peaks of 47 mg m^{-3} at 5 km and 12 km, and summer presents the lowest values of IWC, in accordance with their low frequency in this season. For these two seasons, the largest values of IWC are found in June and October (see right panel of Fig. 12.a), exceeding 50 mg m^{-3} between 5 and 6 km. In general, except in winter, clouds above 10 km have a median ice content ranging from 10 to 40 mg m^{-3}, being greater in fall. For winter, the values of IWC close to the ground were influenced by two snowfall events. Notably, January (see right panel of Fig. 12.a) exhibits much higher IWC near the ground. This is an anomaly attributed to extreme weather events such as the Filomena and Gloria storms that occurred in January of 2020 and 2021, respectively, which significantly impacted the western Mediterranean region (Samuel T. Buisán et al.; María Eugenia Pérez-González et al.; Marta de Alfonso et al.). Normally, lower IWC is observed in this month, but these events contribute to increasing its variability, as can be seen in the bottom panel of Figure 12.a. This panel shows an average IWP much higher than the median in January, indicating lower IWP as typical values, with a median of 5 mg m^{-2}. Overall, the monthly IWP distributions for ice clouds are relatively consistent throughout the year, indicating a well-defined annual IWP pattern, except in January and July. January shows lower occurrence rates and is impacted by the snowfall events, while July is characterized by an extremely low occurrence of ice clouds, resulting in a minimum LWP.

Precipitating ice clouds are the most frequent at our station, with a well defined seasonal cloud occurrence. These clouds are characterized by their greater thickness and lower cloud base height compared to non-precipitating ice clouds. The median profiles of IWC on the left panel of figure 12.c shows similar behavior for summer and fall. Both have a first maximum of 200 mg m^{-3} at 6 km and a second maximum at different altitudes. Summer presents a minor second maxima of about 70 mg m^{-3} around 11 km, and fall has a more pronounced maxima surpassing 200 mg m^{-3} just above 12 km. Moreover, spring and winter have less ice amount and also exhibit similar profiles with only one peak of 100 and 150 mg m^{-3} around 4.5 km, respectively. The right panel of Figure 12.c turns into evidence the area exceeding 50 mg m^{-3}, located between 2 and 7 km over the year. This layer is higher during summer and fall, where values are also larger, achieving more than 200 mg m^{-3} in June and September. In addition, July also achieves these values between 9 and 12 km, different from the observed low IWC for non-precipitating ice clouds. However, mixed-phase clouds and principally precipitating ice mixed phase clouds also exhibited peaks of IWC at this level in summer. These values can be associated with cumulus and cumulonimbus clouds observed in summer at the Granada station. Although this season presents a very limited occurrence of these clouds in summer, they are normally related to convective clouds. Specifically, precipitating ice clouds can be related to cumulonimbus clouds, and mixed-phase and precipitating mixed-phase clouds can be related to cumulus clouds or final stage cumulonimbus clouds, respectively. The bottom panel of Figure 12.c highlights that these clouds contain significantly higher ice water content compared to other clouds, especially during summer and fall, when the 75th percentile increases from 500 to 1000 mg m^{-2}. However, note that despite these values being larger in summer, precipitating ice clouds already showed less cloud occurrence this season, reducing the outliers and turning the IWP values comparable, increasing the 25/75 percentile interval. On the other hand, these clouds have more frequency of occurrence in other months, reducing the variability. It shows that IWP are normally much smaller throughout the year, specially during spring and fall when they are more frequent.

In terms of r_{ice}, the largest sizes for ice clouds can be found below 6 km as can be seen in the left panel of Figure 12.b. Inside this layer, r_{ice} is normally fluctuating around 50 μm, being greater below 3 km only in winter, relating to the snowfalls in January of 2020 and 2021. Above 6 km, ice particle size diminishes, indicating a larger cloud radius at lower levels, a pattern also observed in mixed-phase clouds. Typically, ice particles are larger at low and mid-levels. In the case of cirrus clouds situated above 10 km, the ice effective radius is between 25 and 30 μm, except in winter. These peaks at 11-12 km are coinciding with the IWC. At this layer (10-12 km), the right panel of Figure 12.b display the months responsible for these peaks, which is June, August, October and November (denoted by the white line, contouring regions with more than 30 μm). The monthly distributions in the bottom panel reveals a stable median ice particle size of approximately 38 μm throughout the year, indicating that this value is generally representative when considering all radii within the atmosphere for this cloud type. However, January presents a distribution with a new mode around 50 μm, as expected due to the snowfall events already discussed.

For precipitating ice clouds, the left panel of Figure 12.d shows r_{ice} slightly growing with height above 1 km for all seasons. Below this level, particles are probably related to graupel and are not inside the cloud, then we will consider only the layer above. In general, summer and fall have similar vertical profiles as well as winter has similar profiles as spring up to 10 km, such as the IWC profiles. For winter and spring, r_{ice} increases up to 5 km (reaching 50 μm), whereas in summer and fall, r_{ice} increases up to 7 km (reaching 52 μm). In fact, their maximums are practically the same, but happen at different heights (5 and 7 km). Above their maximums, particle size decreases up to 10 km, when they start to increase again for spring, summer and fall. In spring, r_{ice} starts to increase just above 10 μm and reach at 35 μm at 11 km, while summer maintained r_{ice} at 42 μm between 10 km and 12 km. Then, fall exhibited a sharp increase of r_{ice} at 12 km, reaching a maximum of 65 μm at 13 km. The right panel of Figure 12.d is in agreement with the second maximum of r_{ice} observed in summer, showing the largest particles in July. Also, this panel clearly shows the height where sizes start to decrease with seasons, being higher in summer and fall. The second peaks of spring and fall have not enough neighbor data for painting a contour plot, then they are not shown (but they are still there). At last, despite the larger variability with height, the monthly distribution of r_{ice} in the bottom panel of Figure 12.d shows a highly variable pattern with a sharp mode at the median position of 50 μm.

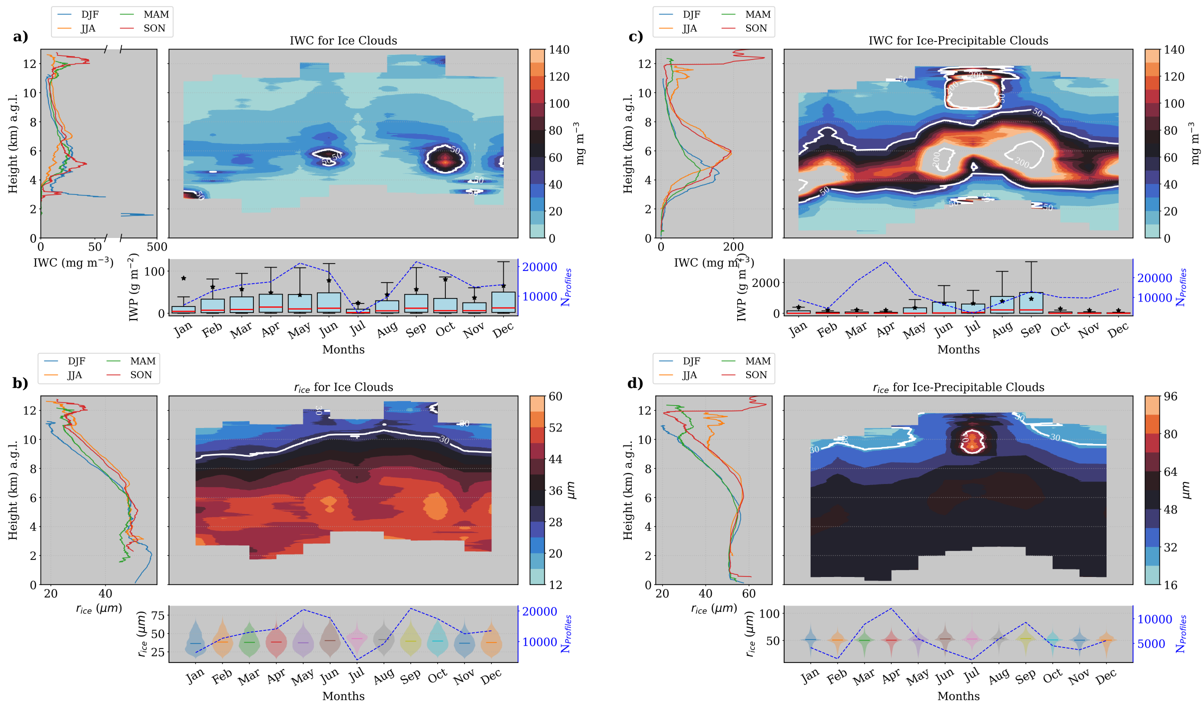

Figure 12. Seasonal and monthly average profiles of microphysical properties, along with monthly IWP and r_{ice}. Panels a) and b) show IWC, r_{ice}, and IWP for non-precipitating ice clouds, while panels c) and d) depict the same parameters for precipitating ice clouds. Specifically: a) and c) display the seasonal (left panel) and monthly (right panel) median profiles of IWC, with boxplots of IWP beneath. Medians are represented by red lines and means by black stars. b) and d) present profiles of r_{ice} for ice and precipitating ice clouds, respectively. The left panels show seasonal medians, while the right panels show monthly medians. The monthly distribution of r_{ice} is illustrated using violin plots in the bottom panel.

In general, non precipitating ice clouds exhibit large amounts of ice water content (IWC) at mid-levels (around 5 km), particularly in early summer and fall. In January, most high IWC values were recorded during extreme snowfall events, such as the Filomena and Gloria storms, resulting in high median IWC at low levels during this month. The r_{ice} is also influenced by these snowfalls, showing higher values close to the surface. Typically, r_{ice} ranges around 50 μm up to 6 km, where it starts to decrease, indicating smaller ice particles in higher-level clouds. Precipitating ice clouds display a much stronger seasonal pattern in their physical properties. Summer and fall show a peak of IWC at 6 km, while winter and spring exhibit a less pronounced peak at 4 km. These clouds contain significantly more IWC than non-precipitating ice clouds, and they are markedly different, with greater thickness. Similar to precipitating mixed-phase clouds, we found large amounts of IWC and r_{ice} at high levels in summer (> 8 km). Additionally, they demonstrate clear seasonal differences in r_{ice}, which is greater in summer and fall from 6 km of height.

6 Conclusions

This study presents a statistical analysis of cloud properties using remote sensing ground-based instruments at the AGORA ACTRIS CCRES station, marking the first such analysis in Spain employing the Cloudnet database. Utilizing over half a decade (2018 to 2023) of post-processed data from the Cloudnet consortium, we classified single-layer clouds and retrieved their physical properties using the target classification product. However, this product exhibited some misclassification at our station, necessitating adjustments to avoid misinterpretation of the results. To enhance cloud classification, we compared previous time-based cloud classification methods from recent studies with a novel approach that categorizes clouds into ice (precipitating ice), liquid (precipitating liquid), and mixed-phase (precipitating mixed-phase) clouds. Evaluating this classification method is crucial for understanding the derived cloud properties, as these will be influenced by the chosen method. Consequently, using this new method, we conducted an inter-annual analysis alongside a diurnal cycle analysis of single-layer clouds to assess seasonal variability and identify convective clouds. Additionally, monthly and seasonal analyses of macrophysical and microphysical properties were presented by cloud type to elucidate their differences.

Appendix A: Cloud assessment by time

This method have been used by recent studies such as Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.. Each study has adopted some different criteria to classify single layer clouds, but they all classify cloud layers by profile. Here, we adopted some criteria based on these studies and employed the following method: cloud layers in the atmospheric column were classified through pixel sequence inquiry. These pixel sequences were divided in the following 5 groups according to Răzvan Pîrloagă et al.: liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only an "ice" pixel sequence was encountered. Mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified whether at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Unlike previous studies, whether rain pixels are found with more than 20 pixels (256-1022 m) of distance from cloud base, the clouds are not classified as precipitable, assuming that rain droplets are not falling from these cloud layers. In all cases, for being considered a cloud layer, a minimum sequence of 3 cloud pixels (38-153 m) (Tatiana Nomokonova et al.) must be satisfied, in addition to a minimum sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" to separate multi-layers. Therefore, free hydrometeor layers in the neighborhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on Răzvan Pîrloagă et al., but these values are not currently well defined, their determination has been guided by empirical experience.

Appendix B: Pixel dilatation

The dilation method consists in expanding the ones in the cloud mask to connect the closest groups of hydrometeors, which are likely to belong to the same cloud. The dilatation follows the scheme below, where hydrometeor pixels are represented in blue. Left image represents the original hydrometeor pixels where two groups of clouds were identified. Thus, a dilatation of 1 pixel is applied in all directions, represented by the red pixels in the right image. Therefore, the dilation process will unite these groups with at least 2 pixels of separation distance. Since they are connected, the cloud classification algorithm will save the coordinates of these cloud pixels in just one group (not two, as previously identified).

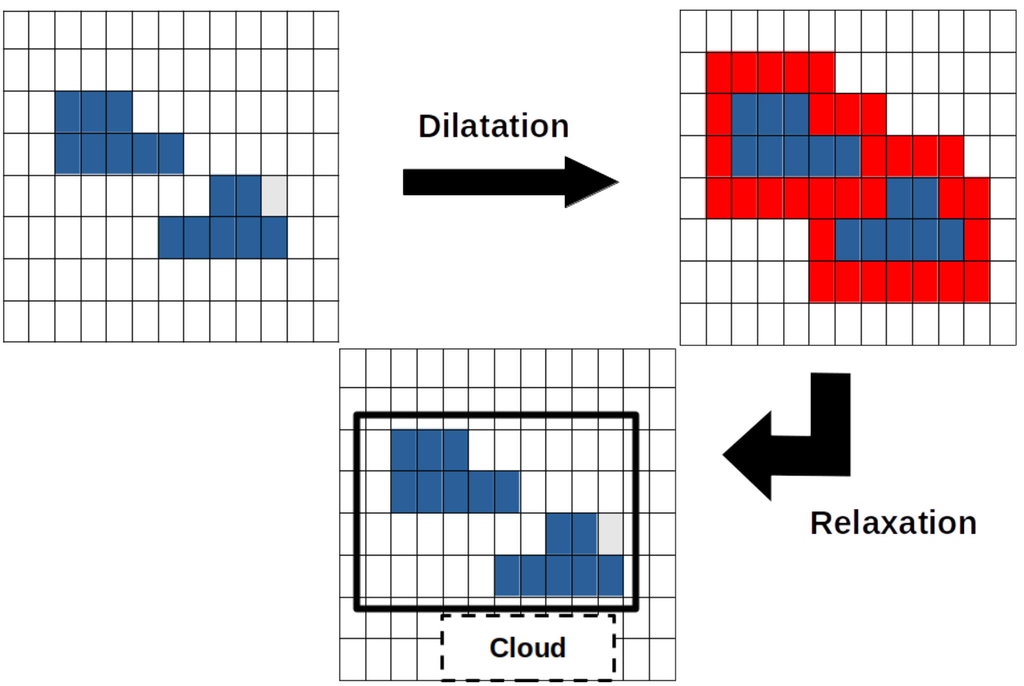

Figure B1. Pixel dilatation and pixel contraction illustration. Left panel shows two hydrometeor groups. Right panel still shows these two clouds and the dilated pixels in red. Since these clouds are connected, the classification algorithm will understand there is only one cloud. Bottom panel shows the result of the pixel contraction, returning cloud pixels to its original position

Next, a contraction process is done to return the coordinates of true hydrometeors to the original position. The contraction will remove the coordinates of fake hydrometeor pixels (pixels in red on the above figure). Afterwards, the classification algorithm can follow its chain and calculate the percentage of each hydrometeor type inside each group to classify different clouds in the cloudnet product.

Acknowledgements. TEXT

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